

MID-TERM EXAMINATION
BTECH (Semester VII)
(October, 2023) OFF LINE mode

Subject Code: BCS 401	Subject: MACHINE LEARNING
Time : 1 ½ Hours	Maximum Marks : 30
Note: Q. 1 is compulsory.	

Q1		(2.5*4)	CO										
	(a) Explain Convergence rule for Machine Learning		CO1										
	(b) Explain Linear Separability in context of Neural Network		CO2										
	(c) For any Machine Learning model differentiate between Bias and Variance		CO1										
	(d) What is the principle of operation of Back Propagation		CO2										
Q2	(Attempt any Two Parts) UNIT-1	(5,5)	CO1										
	(a) What is Gradient Descent Algorithm. Differentiate between Linear and Logistic regression. Give one example of each where these algorithms can be applied.												
	(b) Find Linear Regression equation for following two sets of data. X is an independent variable and Y is a dependent variable. <table><tr><td>X</td><td>2</td><td>4</td><td>6</td><td>8</td></tr><tr><td>Y</td><td>3</td><td>7</td><td>5</td><td>10</td></tr></table> Predict Y for X=4.5	X	2	4	6	8	Y	3	7	5	10		
X	2	4	6	8									
Y	3	7	5	10									
	(c) Explain why is not a good idea to use standard regression algorithms on classification problems. What makes logistic regression algorithm more suitable for classification problems?												
Q3	(Attempt any Two Parts) UNIT-2	(5,5)	CO2										
	(a) What is Curse of Dimensionality. Discuss the need of Dimension reduction in terms of Generalization, Overfitting and Complexity of model												
	(b) Given the data in Table, reduce the dimension from 2 to 1 using the Principal Component Analysis (PCA) algorithm. <table><tr><td>X₁</td><td>4</td><td>8</td><td>13</td><td>7</td></tr><tr><td>X₂</td><td>11</td><td>4</td><td>5</td><td>14</td></tr></table>	X ₁	4	8	13	7	X ₂	11	4	5	14		
X ₁	4	8	13	7									
X ₂	11	4	5	14									
	(c) Explain how linearly inseparable function XOR is implemented using multi layer feed forward neural networks												

MID-TERM EXAMINATION
B.Tech CSE 7th Semester
(October, 2023) OFFLINE mode

Subject Code: BCS 403

Subject: Evolutionary Computing

Time : 1 ½ Hours

Maximum Marks : 30

Note: Q. 1 is compulsory.

Q1		(2.5*4)																																																								
	(a) Explain the need to use Evolutionary Computing (EC). Also explain three areas with examples to apply EC.																																																									
	(b) By briefly describing about Single Arithmetic Recombination (SAR), justify the result of applying SAR on following pair of parents (where $k = 5$, $\alpha = 1/2$):																																																									
	<table border="1" style="display: inline-table; margin: 5px;"> <tr> <td>0.2</td><td>0.5</td><td>0.3</td><td>0.8</td><td>0.6</td><td>0.7</td><td>0.8</td><td>0.5</td><td>0.2</td> </tr> </table> <table border="1" style="display: inline-table; margin: 5px;"> <tr> <td>0.5</td><td>0.7</td><td>0.3</td><td>0.9</td><td>0.2</td><td>0.4</td><td>0.6</td><td>0.3</td><td>0.1</td> </tr> </table>	0.2	0.5	0.3	0.8	0.6	0.7	0.8	0.5	0.2	0.5	0.7	0.3	0.9	0.2	0.4	0.6	0.3	0.1																																							
0.2	0.5	0.3	0.8	0.6	0.7	0.8	0.5	0.2																																																		
0.5	0.7	0.3	0.9	0.2	0.4	0.6	0.3	0.1																																																		
	(c) Design an EA for 8-Queens Problem.																																																									
	(d) Apply n-point crossover on following pair and justify the result of crossover (where $n=3$).																																																									
	1 2 3 4 5 6 7 8 <table border="1" style="display: inline-table; margin: 5px;"> <tr> <td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td> </tr> </table> <table border="1" style="display: inline-table; margin: 5px;"> <tr> <td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td> </tr> </table>	0	1	0	1	1	1	0	1	0	1	0	0	0	0	1	1	0	1																																							
0	1	0	1	1	1	0	1	0																																																		
1	0	0	0	0	1	1	0	1																																																		
Q2	(Attempt any Two Parts) UNIT-1	(5,5)																																																								
	(a) Describe in detail Natural vs. Artificial Evolution. Also, explain the concept of Global and Local optima. (3+2)																																																									
	(b) Implement and discuss in detail the Evolutionary Cycle by Hand for solving the following problem: Maximize $3x^2$ for integers in range 0 – 29. Consider the initial values of x are (12, 25, 19, 7), perform a 1 point crossover and random mutation operations.																																																									
	<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th>S.No</th> <th>x Value</th> <th>Initial population</th> <th>Fitness</th> <th>Prob_i</th> <th>Expected count</th> <th>Actual count</th> </tr> </thead> <tbody> <tr><td>1</td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr><td>2</td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr><td>3</td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr><td>4</td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr><td>Sum</td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr><td>Avg</td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr><td>Max</td><td></td><td></td><td></td><td></td><td></td><td></td></tr> </tbody> </table>	S.No	x Value	Initial population	Fitness	Prob _i	Expected count	Actual count	1							2							3							4							Sum							Avg							Max							
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	(c) Explain the general scheme of an evolutionary algorithm in detail. Also, explain its components. (2+3)																																																									
Q3	(Attempt any Two Parts) UNIT-2	(5,5)																																																								
	(a) Write in detail about Edge crossover. Apply Edge crossover using the Edge table on																																																									

	the following parents and discuss the results. Parent 1 <table border="1"><tr><td>8</td><td>7</td><td>4</td><td>5</td><td>6</td><td>9</td><td>3</td><td>2</td><td>1</td></tr></table> Parent 2 <table border="1"><tr><td>6</td><td>5</td><td>1</td><td>3</td><td>2</td><td>7</td><td>8</td><td>4</td><td>9</td></tr></table>	8	7	4	5	6	9	3	2	1	6	5	1	3	2	7	8	4	9																			
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6	5	1	3	2	7	8	4	9																														
(b)	Describe various types of Permutation mutation. Apply Swap mutation on parent 1 & parent 2, apply Insert mutation on parent 3 & parent 4, and discuss the result in detail: (3+2) Parent 1 <table border="1"><tr><td>8</td><td>7</td><td>3</td><td>5</td><td>6</td><td>9</td><td>4</td><td>2</td><td>1</td></tr></table> Parent 2 <table border="1"><tr><td>1</td><td>2</td><td>6</td><td>7</td><td>8</td><td>3</td><td>4</td><td>5</td><td>9</td></tr></table> Parent 3 <table border="1"><tr><td>5</td><td>6</td><td>1</td><td>3</td><td>2</td><td>7</td><td>8</td><td>4</td><td>9</td></tr></table> Parent 4 <table border="1"><tr><td>2</td><td>3</td><td>9</td><td>6</td><td>7</td><td>8</td><td>1</td><td>4</td><td>5</td></tr></table>	8	7	3	5	6	9	4	2	1	1	2	6	7	8	3	4	5	9	5	6	1	3	2	7	8	4	9	2	3	9	6	7	8	1	4	5	
8	7	3	5	6	9	4	2	1																														
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2	3	9	6	7	8	1	4	5																														
(c)	Explain tree representation along with an example. Also, perform recombination on the example selected. (Take assumptions, wherever necessary) (2.5+2.5)																																					

(Please write your Enrollment Number)

Enrollment No. 001

(Please write your Enrollment Number)

Enrollment No. 1501

MID-TERM EXAMINATION
(CBCS/Non CBCS)

Course Name: < B.Tech CSE/IT >, Semester: < 7th >
(October, 2023)

Subject Code: BIT 401	Subject: Mobile Communications
Time : 1 ½ Hours	Maximum Marks : 30
Note: Q. 1 is compulsory. Attempt any two questions from the rest.	

Q1		(2*5)	
(a)	What is the uplink frequency band in UMTS networks	0.5	
(b)	Enumerate the components of an GSM network	1	
(c)	What is PAN?	2	
(d)	What is frequency hopping?	2	
(e)	Why are cells assumed to be hexagonal?	2	
Q2		(5,5)	
(a)	Explain GPRS architecture with relevant diagrams	✓	
(b)	Explain the various Traffic and Control Channels in UMTS	2.5	
Q3		(5,5)	
(a)	Explain the concept of Handover in an UMTS system.		
(b)	What is the motivation for a specialized MAC		
Q4		(5,5)	
(a)	Explain CDMA with a relevant example	3	
(b)	Explain the various signal propagation effects	4	

MID-TERM EXAMINATION
(Course Name : B.Tech IT) (Semester 7th)
(October, 2023) OFF LINE mode

Subject Code: BIT 417	Subject: E-commerce
Time : 1 ½ Hours	Maximum Marks : 30
Note: Q. 1 is compulsory.	

Q1		(2.5*4)	
	(a) Give some potential benefits of E-commerce.		
	(b) Explain various initiatives taken by the government to help grow e-commerce.		
	(c) Explain the features of E-Business.		
	(d) Define CRM and its types.		
Q2	(Attempt any Two Parts) UNIT-1	(5,5)	
	(a) Write the difference between traditional commerce and e-commerce with an example.		
	(b) Explain different types of e-commerce models with a real-world example.		
	(c) What are the forces behind the growth of e-commerce?		
Q3	(Attempt any Two Parts) UNIT-2	(5,5)	
	(a) What are the requirements that can help in growing the E-Business?		
	(b) What are the factors behind the success of e-business?		
	(c) Explain ERP and its benefits with a case study.		

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(Offline)
Course Name: <B.TechCSE>, Semester: <7th>
(November-December, 2023)

Subject Code: BIT 417	Subject: E-Commerce	Maximum Marks :60
Time :3 Hours		
Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.		

Q1		(2.5*8=20)	
(a)	Explain the components of e-commerce.		
(b)	What are the Incentives for engaging in electronic commerce?		
(c)	Explain the requirements of e-business.		
(d)	Differentiate between E-business and E-commerce.		
(e)	Explain micropayments and security issues in e-commerce.		
(f)	What are the benefits of VPN connection?		
(g)	Explain the features of LANs.		
(h)	Explain different players in m-commerce.		
UNIT-I			
Q2	Explain the architectural framework of e-commerce.	(10)	
Q3	(a) Explain the advantages and disadvantages of e-commerce. (b) Explain various technological trends in E-commerce.	(5,5)	
UNIT-II			
Q4	Explain MRP and its process with a diagram. How it is different from ERP.	(10)	
Q5	(a) Explain the level of e-business strategy. (b) Explain the types of positioning strategies	(5,5)	
UNIT-III			
Q6	(a) Explain various e-commerce laws. (b) Explain the process of a secure payment system.	(5,5)	
Q7	(a) Explain the firewall and its characteristics. (b) Explain various encryption tools.	(5,5)	
UNIT-IV			
Q8	Explain the architecture and stack of Bluetooth with a real-time application of it.	(10)	
Q9	(a) Advantages & Disadvantages of M-commerce. (b) Differentiate between m-commerce and e-commerce with one real-world example.	(5,5)	

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:<B.Tech CSE/ ECE>, Semester:<7th>
(November-December, 2023)

Subject Code: BCS 401	Subject: Machine learning
Time :3 Hours	Maximum Marks :60
Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.	

Q1		(2.5*8=20)																						
	(a) Compare Active and Passive Learner with example <u>1</u>																							
	(b) Explain the need of Dimension Reduction. Give name of two methods for dimension reduction																							
	(c) Can logistic regression use for more than 2 classes? Justify your answer <u>0</u>																							
	(d) Consider the data to classify whether given piece of cloth is Good or Bad. Now a company produces a new cloth that pass lab test with $X_1=3$ and $X_2=7$. Classify this cloth in good or bad category by using KNN.																							
	<table border="1" style="margin-left: auto; margin-right: auto;"> <tr> <th>X_1</th> <th>X_2</th> <th>Y</th> </tr> <tr> <td>7</td> <td>7</td> <td>Bad</td> </tr> <tr> <td>7</td> <td>4</td> <td>Bad</td> </tr> <tr> <td>3</td> <td>4</td> <td>Good</td> </tr> <tr> <td>1</td> <td>4</td> <td>Good</td> </tr> </table> <u>2</u> <u>11.5 + 2</u> <u>13.5</u>	X_1	X_2	Y	7	7	Bad	7	4	Bad	3	4	Good	1	4	Good								
X_1	X_2	Y																						
7	7	Bad																						
7	4	Bad																						
3	4	Good																						
1	4	Good																						
	(e) Why is "Naive" Bayes naïve?																							
	(f) What is Cluster analysis? How is it different from Classification?																							
	(g) Explain neural network approach to clustering? <u>1</u>																							
	(h) Compare Forward and Backward error Propagation in Neural Network																							
	UNIT-I	(5,5)																						
Q2	(a) Sales data of 10 months for a coffee house situated near a prime location of a city comprising the number of customers (in hundreds) and monthly sales (in Thousand Rupees) are given below: Find the simple linear regression equation that fits the given data.																							
	<table border="1" style="margin-left: auto; margin-right: auto;"> <tr> <th>No. of Customers (in hundreds)</th> <th>Monthly Sales (in thousand Rs.)</th> </tr> <tr><td>6.0</td><td>01</td></tr> <tr><td>6.1</td><td>06</td></tr> <tr><td>6.2</td><td>08</td></tr> <tr><td>6.3</td><td>10</td></tr> <tr><td>6.5</td><td>11</td></tr> <tr><td>7.1</td><td>20</td></tr> <tr><td>7.6</td><td>21</td></tr> <tr><td>7.8</td><td>22</td></tr> <tr><td>8.0</td><td>23</td></tr> <tr><td>8.1</td><td>25</td></tr> </table> <u>41.5</u> <u>42</u> <u>2</u> <u>44</u> <u>41.5</u> <u>2</u> <u>43.5</u> <u>1.5</u> <u>45</u>	No. of Customers (in hundreds)	Monthly Sales (in thousand Rs.)	6.0	01	6.1	06	6.2	08	6.3	10	6.5	11	7.1	20	7.6	21	7.8	22	8.0	23	8.1	25	
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7.6	21																							
7.8	22																							
8.0	23																							
8.1	25																							

(b) Derive Cost function for Linear Regression. What are the four assumptions of linear regression (simple linear and multiple)?

- Q3 (a) What is the difference between Maximum likelihood estimation and Gradient descent.
(b) Define following performance metrics for any ML model: Accuracy, Precision, ROC Curve, Confusion matrix, F1 score.

UNIT-II

Q4 For given data Calculate PCA

(10)

2	1
3	4
5	0
7	6
9	2

Q5 Consider 2D pattern as follows (1,1), (1,2), (4,4), (5,4) Use PCA to find 1D data.

UNIT-III

- Q6 (a) Name two decision tree induction algorithms. How these algorithms are different from each other?
(b) Using the given table, show how the induction of a decision tree is done using information gain

(5,5)

Attributes				
Outlook	Temperature	Humidity	Windy	Class
Sunny	Hot	High	False	N
Sunny	Hot	High	True	N
Overcast	Hot	High	False	P
Rain	Mild	High	False	P
Rain	Cool	Normal	False	P
Rain	Cool	Normal	True	N
Overcast	Cool	Normal	True	P
Sunny	Mild	High	False	N
Sunny	Cool	Normal	False	P
Rain	Mild	Normal	False	P
Sunny	Mild	Normal	True	P
Overcast	Mild	High	True	P
Overcast	Hot	Normal	False	P
Rain	Mild	High	True	N

Q7

(a) Discuss following strategies for improving the classifier and predictor accuracy: i) Bagging ii) Boosting.

(b) Describe learning with Naïve Bayes model. Consider the following attributes and classify whether the car will be stolen or not:

Example	Origin	Type	Color	Stolen
1.	Domestic	Sports	Red	Yes ✓
2.	Domestic	Sports	Red	No
3.	Domestic	Sports	Red	Yes ✓
4.	Domestic	Sports	Yellow	No
5.	Domestic ✓	SUV	Yellow	No
6.	Imported	Sports	Yellow	Yes ✓
7.	Imported	Sports	Yellow	No
8.	Imported	SUV	Red	No
9.	Imported	SUV	Yellow	Yes ✓
10.	Imported	SUV	Yellow	No

UNIT-IV

(5,5)

Q8

(a) Discuss different type of clustering methods

(b) Consider the following sample of 2D data sample. Apply k means clustering for k=2.

Age	Income(in thousands)
20	10
30	20
30	30
35	35
40	40
50	45

Q9

(a) What is cluster analysis how it is different from classification? Give four examples of clustering problem

(b) Apply K means clustering for following data. Assume $C_1(2)$, $C_2(16)$ and $C_3(38)$ as initial clusters: 2,4,6,3,31,12,15,16,38,35,14,21,23,25,30

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(Offline)
Course Name: B.Tech CSE, Semester: 7th Sem
(November-December, 2023)

Subject Code: BCS 403

Subject: Evolutionary Computing

Time : 3 Hours

Maximum Marks : 60

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(2.5*8 =20)																				
2.5	(a) Write Pseudo code for EA Scheme.																					
2.5	(b) Perform Simple Arithmetic crossover on below data with $k=4$ and $\alpha = 0.3$.																					
2.5	<table border="1" style="margin: 10px auto; border-collapse: collapse;"> <tr> <td style="padding: 2px 10px;">0.5</td> <td style="padding: 2px 10px;">0.1</td> <td style="padding: 2px 10px;">0.6</td> <td style="padding: 2px 10px;">0.4</td> <td style="padding: 2px 10px;">0.6</td> <td style="padding: 2px 10px;">0.8</td> <td style="padding: 2px 10px;">0.4</td> <td style="padding: 2px 10px;">0.2</td> <td style="padding: 2px 10px;">0.9</td> </tr> </table>	0.5	0.1	0.6	0.4	0.6	0.8	0.4	0.2	0.9												
0.5	0.1	0.6	0.4	0.6	0.8	0.4	0.2	0.9														
2.5	(c) Differentiate between Genetic algorithm and Genetic programming																					
1.5	(d) Describe how ants are able to find the shortest path to a food source.																					
1.5	(e) What are variation operators? Explain with an example.																					
1.5	(f) Perform cycle crossover on the below parents																					
1.5	<table border="1" style="margin: 10px auto; border-collapse: collapse;"> <tr> <td style="padding: 2px 10px;">P1:</td> <td style="padding: 2px 10px;">9</td> <td style="padding: 2px 10px;">3</td> <td style="padding: 2px 10px;">7</td> <td style="padding: 2px 10px;">8</td> <td style="padding: 2px 10px;">2</td> <td style="padding: 2px 10px;">6</td> <td style="padding: 2px 10px;">5</td> <td style="padding: 2px 10px;">1</td> <td style="padding: 2px 10px;">4</td> </tr> <tr> <td style="padding: 2px 10px;">P2:</td> <td style="padding: 2px 10px;">1</td> <td style="padding: 2px 10px;">2</td> <td style="padding: 2px 10px;">3</td> <td style="padding: 2px 10px;">4</td> <td style="padding: 2px 10px;">5</td> <td style="padding: 2px 10px;">6</td> <td style="padding: 2px 10px;">7</td> <td style="padding: 2px 10px;">8</td> <td style="padding: 2px 10px;">9</td> </tr> </table>	P1:	9	3	7	8	2	6	5	1	4	P2:	1	2	3	4	5	6	7	8	9	
P1:	9	3	7	8	2	6	5	1	4													
P2:	1	2	3	4	5	6	7	8	9													
1.5	(g) What is bloating in Genetic Programming? OR Explain the initialization process of Genetic Programming.																					
2.5	(h) What are the limitations of Swarm Intelligence?																					
UNIT-I																						
Q2	Diagrammatically explain all stages of typical EA behavior. Differentiate between Natural and Artificial Evolution (4+6)	(10)																				
Q3	Represent, using pseudo code, the Evolutionary Algorithm. Discuss the main components used to solve an EA. (4+6)	(10)																				
UNIT-II																						
Q4	Assume we have the following function: $\max f(x) = x^3 - 60 * x^2 + 900 * x + 100$. Using a binary representation, we can represent x using five binary digits. Given the following four chromosomes, give the values for x and $f(x)$. <table border="1" style="margin: 10px auto; border-collapse: collapse;"> <thead> <tr> <th style="padding: 2px 10px;">Chromosome</th> <th style="padding: 2px 10px;">Binary String</th> </tr> </thead> <tbody> <tr> <td style="padding: 2px 10px;">P₁</td> <td style="padding: 2px 10px;">11100</td> </tr> <tr> <td style="padding: 2px 10px;">P₂</td> <td style="padding: 2px 10px;">01111</td> </tr> <tr> <td style="padding: 2px 10px;">P₃</td> <td style="padding: 2px 10px;">10111</td> </tr> <tr> <td style="padding: 2px 10px;">P₄</td> <td style="padding: 2px 10px;">00100</td> </tr> </tbody> </table> Also, perform n-point crossover on the fittest individual using $n=4$. (8+2)	Chromosome	Binary String	P ₁	11100	P ₂	01111	P ₃	10111	P ₄	00100	(10)										
Chromosome	Binary String																					
P ₁	11100																					
P ₂	01111																					
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P ₄	00100																					
Q5	Suppose a genetic algorithm uses chromosomes of the form $x: a b c d e f g h$ with a fixed length of eight genes. Each gene can be any digit between 0 and 9. Let the fitness of individual x be calculated as: $f(x) : (a + b) - (c + d) + (e + f) - (g + h)$, and let the initial population consist of four individuals with the following chromosomes:	(10)																				

	$x1 = 65413532$ $x2 = 87126601$ $x3 = 23921285$ $x4 = 41852094$ Evaluate each individual's fitness and arrange them in order, with the fittest first and the least fit last. Perform the following crossover operations: i) Cross the fittest two individuals using a one-point crossover at the middle point. ii) Cross the second and third fittest individuals using a two-point crossover (points b and f). (6+2+2)		
	UNIT-III		
	Q6	Describe the PSO Algorithm in detail along with the pseudo code. Also, explain how the velocity and position of particles are updated. (6+4)	(10)
10	Q7	Discuss various popular Evolutionary Algorithm variants. (all equal marks)	(10)
	UNIT-IV		
10	Q8	Write about the Firefly algorithm (Behavior, Characteristics, Steps, and Algorithm). Also, explain four application areas of the firefly algorithm. (2*5 = 10)	(10)
	Q9	Explain Ant Colony Metaheuristics. Write about three application algorithms of Ant Colony Optimization. (4+6)	(10)

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
B.Tech(IT/CSE), Semester:<7th>
(November-December, 2023)

Subject Code: BIT 401	Subject: Mobile Computing
Time :3 Hours	Maximum Marks :60
Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.	

Q1		(2.5*8=20)	
(a)	What is the ISM band?		
(b)	What are satellite phones? What is there utility?		
(c)	Explain network controlled handoff		
(d)	Draw the UMTS architecture		
(e)	What is tunneling?		
(f)	Explain fast retransmit briefly		
(g)	Why are blackberry devices not used anymore?		
(h)	Compare HiperLan1 and HiperLan2 on any 4 valid points		
UNIT-I			
Q2	Discuss various channel assignment strategies	(10)	
Q3	Compare TDMA and CDMA with relevant examples. Also enumerate similarities if any?	(10)	
UNIT-II			
Q4	Explain GPRS network in detail with relevant diagrams	(10)	
Q5	Explain the reasons for a specialized MAC in wireless networks	(10)	
UNIT-III			
Q6	Mention various types of agents used in Mobile IP. Explain the connection establishment between the mobile node in home network and corresponding node in foreign network. Draw relevant diagrams where required	(10)	
Q7	What is routing .Explain DSDV with a relevant example	(10)	
UNIT-IV			
Q8	Draw the architecture of WAP protocol stack. Give the functionality of each layer.	(10)	
Q9	Explain Bluetooth protocol stack with relevant diagrams	(10)	

10 12

7 8.5

6

8.5